# Changing a Networked Mind: Paradigm Dynamics as Structure Learning over a Hidden World

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**Abstract.** How does a scientific community change its mind—not about a fact, but about the *frame* its facts live in? We model a paradigm as a directed network of commitments and a community as bounded active-inference agents who must *learn* that network's structure while valuing some conclusions over others and pooling beliefs with peers. Driving this community through the full Kuhnian cycle, one signature organizes everything we find: *conflict resolves into confident compromise.* Valuing a conclusion shifts where beliefs settle but never delays their yielding; under any contact, communities converge; the lone discoverer of a new dimension is averaged away. These are theorems of the regime, not accidents of parameters—Gaussian beliefs under precision-weighted pooling can neither reject a conflicting source nor sustain a school—and they locate what paradigm change requires: value entering the *precision* of beliefs rather than their means, and pooling that reads *content.* Averaging refines a frame; it cannot grow a new one.

**Keywords:** Active inference · Structure learning · Belief sharing · Paradigm change · Social epistemology

## 1 Introduction: changing a frame, not a credence

If we want science to keep going well, goodwill is not enough: we need to know how science *works*—what kind of agent can do science at all, how a community comes to represent a genuinely new dimension of the world, and how it loses information it once held. The question has lately acquired a second audience, because communities of reasoning agents are no longer only studied but *built*—machine-mediated research pipelines, automated literature synthesis, multi-agent learning systems—and every such community commits, in its aggregation step, to an answer it rarely examines: what happens to the member who disagrees. Machine learning has begun to name the property at stake *open-endedness*, the capacity of a system to keep producing artifacts that are novel and learnable, identified both as a grand challenge [28] and as an essential ingredient of any general intelligence [14]; social epistemology has long catalogued its failure modes, echo chambers [23] and premature consensus [32]. What neither supplies is a foundation: a model of a *community* of structure-learning agents explicit enough that one can prove which aggregation dynamics preserve open-endedness and which foreclose it. This paper builds that foundation, and its central result is a negative one with constructive content: the most natural baseline—Gaussian beliefs under precision-weighted pooling—provably lacks the capacity. Averaging refines a frame; it cannot grow a new one.

To change one's mind about a fact is to update a belief; to change a paradigm is to change the *frame* in which beliefs are posed. A frame is not a credence but a *structure*: a network of conditional dependencies through which information propagates. A community that revised only its credences, leaving the network fixed, could never change frame at all. This is the phenomenon Kuhn named but could not run forward [19]: nothing in *The Structure of Scientific Revolutions* says, mechanistically, why a community holds a structure of commitments long past the first disconfirming evidence, nor what finally makes the holding give way.

When the chemical community gave up phlogiston it did not revise a single credence. It rewrote a network of commitments in which combustion, calcination, respiration, and the theory of acids were all load-bearing on a hub that had to go: late-phlogistic theorists proposed cascading revisions that proved mutually incompatible [40, 43], defending less a hypothesis about combustion than a constellation of entangled commitments [42]. Phlogiston was load-bearing for too much.

Network epistemology, the dominant formalism for such communities, cannot represent this case. Its agents hold a scalar credence in a fixed proposition and update it on a trust graph [24, 32, 75]—a programme with striking results on transient diversity, polarization, and the division of cognitive labour [16, 17, 29, 45], but the phlogiston puzzle is not a failure to converge on a number. The commitments were *entangled*: revising one forced a re-reading of the others, and the dispute was over which entanglements to keep. Thagard's ECHO [30, 41] captures the structural intuition as undirected coherence, but lacks the directed dependencies, the carry-over cost of mind-change, and the expand/reduce asymmetry we derive below. Active-inference accounts supply an endogenous, precision-based lock-in mechanism [1] and warn that naive aggregation degrades collective precision [47, 48]; the variational cost of mind-change has been formalized for a single agent as a motivated trade between revision cost and the utility of a held belief [15]. What none of these does is make *belief structure itself* the object of inference. A scalar credence over a fixed menu of hypotheses cannot say how revising one commitment forces a re-reading of the rest, nor how a community arrives at a *new* way of carving up the world: its members can herd on a number or split over it, but they can never come to hold different *kinds* of world—only different values on a shared one.

This paper lets a piece of *structure* travel, so that one agent can come to represent a commitment its peers do not. Its agents carry a directed network of commitments and learn that network by Bayesian model expansion and reduction; they are *bounded*, sampling a complex world through narrow channels, valuing some conclusions over others, and judging what they receive against their own model. Section 3 assembles this community from the field's own default choices; Section 4 drives it through the full Kuhnian cycle—normal science, anomaly, crisis, reduction, discovery—and opens with why one signature had to organize everything found there: two classical theorems under which the regime can neither reject a conflicting source nor sustain a school, so that *conflict resolves into confident compromise*. What the sweep measures is where that regime ends. Conviction moves where beliefs settle, never when they yield; what protects an old paradigm is isolation, and what kills a new concept is company. The discussion draws the constructive map: which mechanisms paradigm dynamics actually require, and where they must enter. The story, compactly, is about averaging—what belief revision is when value enters as a tilt rather than a stiffness, what propagation-by-averaging does to a population of structure learners, and what it takes for multi-agent structure learning to be *open-ended* rather than self-sealing.

## 2 What a model of science must contain

Three things, at least. *Structure as the object of inference*: a scientist confronting a new domain is not given the right variables, let alone the dependencies among them—whether two regularities share a hidden common cause, are lawfully coupled, or merely coincide is exactly what she must learn. Frames differ in kind, not in degree, so credences over a fixed menu cannot carry them; the theory-acquisition literature reaches the same conclusion, casting conceptual development as a stochastic search over structures [34, 35, 49]. Once structure is the object, exploration stops being a hand-tuned bonus—an $\epsilon$, a temperature, a switching cost [24, 32]—and becomes the epistemic value of *expanding* the model to entertain a commitment not yet in it [9, 10], with Bayesian model reduction as its complement, pruning what the data abandon in closed form [11, 50]. Existing applications of that machinery run on single agents [50, 51] or define "group" as between-subject neuroimaging [52]; a *community* jointly expanding and reducing a paradigm is the open move this paper makes.

*Value*, because scientists care about their conclusions: a result on which a career rests is held more tightly than its evidence warrants [17, 29], and attitudes demonstrably live on networks of exactly this kind [3]. And *communication*, because science is social: beliefs travel a trust graph and must somehow be combined when they meet. Each requirement has a default formalization, so natural it is hardly ever examined. Beliefs over a network become a precision-weighted Gaussian model; value becomes a utility that tilts the posterior [15]; combination becomes precision-weighted averaging, the information form of classical opinion pooling [60, 62]. Section 3 builds the community out of exactly these defaults, on purpose. The question the rest of the paper asks is whether such a community can ever do more than *refine*: whether agents built this way can grow a frame, and not merely polish the one they hold.

## 3 The model

We draw the system whole first—a population of agents on a trust graph—then build the single agent up from first principles, and finish by stating the dynamics as an explicit per-round loop (§3.5).

### 3.1 The population, and the paradigm network each agent carries

The third requirement of §2—communication—fixes the shape of the whole, so we draw the whole first. A population of $N$ agents inhabits a shared hidden world. Agents are coupled on a trust graph: communities are blocks of a stochastic block model, with dense intra-community edges and an inter-community coupling weight (*inter*) that we sweep from total isolation to dense contact. Each agent privately carries its own *paradigm network*, a directed graph $G = (V, E, \Pi)$: nodes carry theoretical commitments $x_v$, edges carry inferential dependencies, and each edge bears a coupling precision $\pi_{uv}$ measuring how tightly $v$ is bound to $u$—from which the agent's conservatism and conviction will be derived (Eqs. 1 and 2 below). Figure 1 draws the picture.

Two kinds of things propagate on this underlying network. Beliefs and accumulated precisions are fused across trust edges in information form—a precision-weighted pooling of posteriors in which an agent's own evidence and its neighbours' carry weight in proportion to trust, the information-form counterpart of trust-weighted averaging in opinion dynamics [55, 60]. And—this part is not standard—*structure* spreads the same way: an agent whose neighbour has awakened a new node receives the proposal and scores it against its own model (§3.3). What travels on the trust graph is not only a number but a piece of structure, evaluated against wiring the sender does not see.

What does one of those private networks hold? Begin concretely. The phlogiston paradigm, at its smallest, is three commitments wired together: combustion releases phlogiston; calcination is combustion, slowed; respiration is combustion, tamed. Each commitment conditions the next, evidence for any one bears on all, and the question "what if the calx *gains* weight?" has no local answer—it tugs the whole web. A paradigm, in this paper, is that web made explicit.

The network is just a generative model whose hidden states are the commitments themselves, with precision playing its usual role: a high-$\pi$ edge says that belief in $u$ strongly constrains belief in $v$. Internal coherence is the corresponding precision-weighted graphical model, $p(x \mid G) \propto \prod_{(u \to v) \in E} \psi_{uv}(x_u, x_v)^{\pi_{uv}}$. None of this machinery is ours: it is a linear-Gaussian structural equation model in the lineage of Wright's path analysis [74], edges as path coefficients and precisions as their weights. We work throughout in that *linear-Gaussian* regime, which puts every quantity below in closed form (Appendix A); agents are *honest*—no one misreports, so lock-in is never a dishonesty artifact—and *bounded*, minimizing variational free energy step by step [39] rather than planning over policies.

Revising a commitment is not a local act: changing $x_v$ forces everything downstream to re-equilibrate, along every chain of dependence. If $A$ is the matrix of edge precisions, $A^k$ collects the chains of length $k$, and summing chains of every

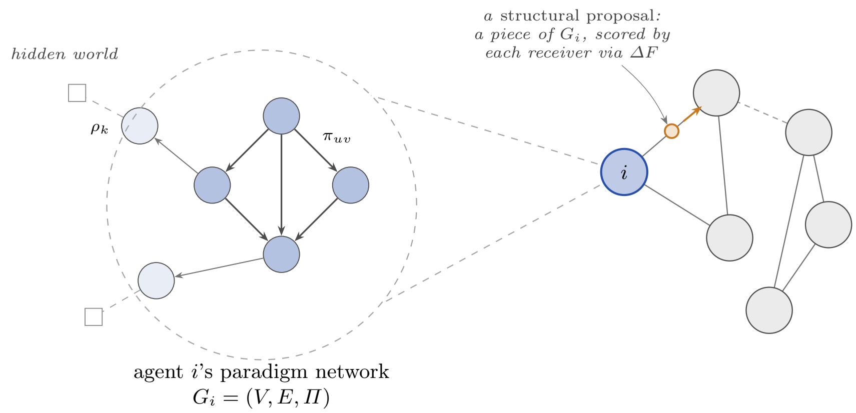



Fig. 1: The population model. Right: the community is a trust graph, every circle an agent. Left, magnified: each agent privately holds a paradigm network $G_i = (V, E, \Pi)$—commitments coupled by precisions $\pi_{uv}$, a belt reading the hidden world through sensory channels of precision $\rho_k$—from which its conservatism $\kappa = \mathsf{T}\mathbf{1}$ and conviction $U = \mathsf{T}u$ are derived. What passes along trust channels (amber) is a *structural proposal*: a piece of paradigm, e.g. a newly awakened node, which each receiver scores against its own network by $\Delta F$ (Eq. 5); posteriors fuse precision-weighted along the same channels. The trust graph is fixed at the start (a stochastic block model): channels are dense and strong inside a community, and a single weak weight *inter*—the main dial of §4—connects communities, so what reaches an agent from outside is its neighbours' evidence and proposals, scaled by trust.

length gives $\sum_k A^k = (I - A)^{-1}$—a finite sum on a DAG. Two literatures already own this object: path analysis as the *total-effects matrix* [58, 74], network science as the Katz–Bonacich index [59, 64]; and it is exactly how influence travels between beliefs in precision-weighted Gaussian models [70]. We adopt it as the *propagation operator* $\mathsf{T} = (I - A)^{-1}$, indexed with rows as sources and columns as descendants, so the entry $\mathsf{T}_{iw} = \pi_{i \rightsquigarrow w}$ is the total precision-weighted coupling propagated from $i$ to $w$ along all directed paths (Appendix A).

Conservatism is then recognized rather than posited. Place a unit of weight on every node and ask, for each commitment $i$, how much total weight a revision of $i$ would disturb. That is the operator applied to the unit source,

$$\kappa \;=\; \mathsf{T}\mathbf{1}, \qquad \kappa_i \;=\; 1 + \sum_{w \in \mathrm{desc}(i)} \pi_{i \rightsquigarrow w}, \tag{1}$$

the precision-weighted mass of a node's descendants. In network-science terms this is a centrality score—a node scores high when many strong chains of dependence leave it (formally, the broadcast variant of the Katz index)—and centrality is usually read as influence or status. We read the same number as *revision cost*: a commitment is exactly as expensive to revise as the mass that must re-equilibrate when it moves. Conservatism is thus a centrality the graph already determines, not a disposition assigned by hand.

What we add to this inherited machine comes in two layers. The first is contact with the world. The dense, high-precision interior of the network is the *core*; the sparse periphery in contact with observation is the *belt*, each belt node reading the world through a sensory channel of precision $\rho_k$. The core carries no direct sensory channel: an interior commitment is revised only by error propagated inward from the belt, through the same operator—costly by Eq. (1) and insulated by position, which is what makes it slow.

The second layer is value, and the operator is run twice. Scientists value some conclusions over others (§2); following the variational-cost account of belief revision [15], we attach to each node an intrinsic utility $u_i$ and let it propagate through the *same* operator: a commitment one wants true lends value to its dependents, so the conviction field is

$$U \;=\; \mathsf{T}\, u, \qquad U_i \;=\; u_i + \sum_{w \in \mathrm{desc}(i)} \pi_{i \rightsquigarrow w}\, u_w. \tag{2}$$

In the operator's path-analytic reading, $U$ is the total effect of wanting things true. Where the inherited models propagate a single source, a paradigm carries two: $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$ are linearly independent unless $u \propto \mathbf{1}$, so the two fields need not align (Fig. 2).

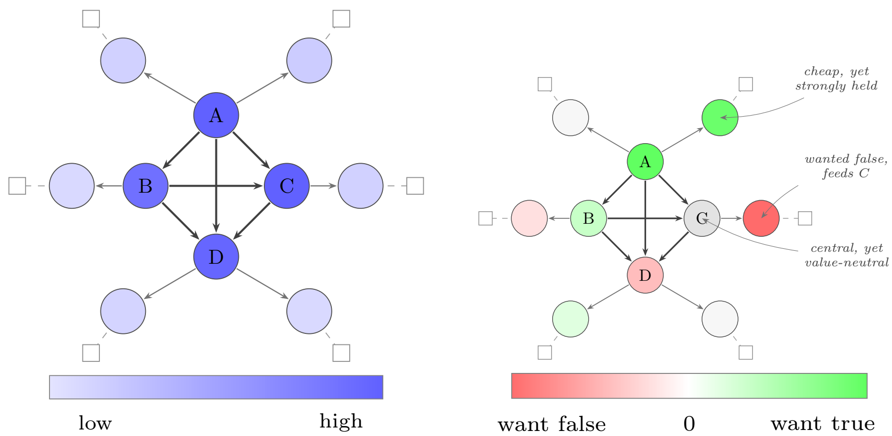
### 3.2 Motivated revision and the lock-in parameter

The bounded agent revises its belief $q(x)$ over the whole network by trading fidelity to the honest posterior against the value of where the commitments land,

$$\mathcal{F}[q] \;=\; D_{\mathrm{KL}}[q(x)\,\|\,p(x \mid o, G)] - \lambda\, \mathbb{E}_q\big[U^{\top} x\big]\,, \qquad \Longrightarrow \qquad q_\lambda(x) \;\propto\; p(x \mid o, G)\, e^{\lambda\, U^{\top} x}, \tag{3}$$

with $\lambda$ a single temperature governing how far value may bend the evidence. The operation itself is classical—an exponential tilt of the posterior, the form KL-regularized utility maximization always yields [72, 73]; what is ours is the source of the tilt, a conviction field propagated through the paradigm's own operator. In the linear-Gaussian regime the tilt is a pure mean shift, displacing the commitments by $\lambda \Sigma U$ and leaving the covariance untouched (Appendix A). What opposes the displacement is evidence, also through the mean: a disconfirming channel $k$ with sensory precision $\rho_k$ and read-out $H_k$ pulls the mean back with gain $\propto \rho_k\, H_k \Sigma U$. As conviction drives a disconfirming channel toward $\rho_k \to 0$ that gain vanishes: the channel can no longer move the mean, and the value-displaced commitment stands unrefuted. The fraction of disconfirming sensory precision so silenced,

$$\gamma \;=\; \frac{\sum_{k \in \mathrm{dis}} \rho_k\, \mathbb{1}[\,\rho_k \text{ gated by the core}\,]}{\sum_{k \in \mathrm{dis}} \rho_k}, \tag{4}$$




(a) Revision cost $\kappa_i$ (conservatism) (b) Intrinsic utility $u_i$ (conviction)

Fig. 2: Two fields on one paradigm (schematic). Both panels carry the *same* network—a densely coupled core $\{A, B, C, D\}$ and a sparse belt $\{b_1, \ldots, b_6\}$ in contact with observation. a Revision cost is the carry-over of mind-change, $\kappa = \mathsf{T}\mathbf{1}$ (Eq. 1): a node's cost is the precision-weighted mass of its descendants, a clean radial gradient monotone in centrality. b Intrinsic utility is signed—beliefs one wants true (green), wants false (red), or is neutral toward (white)—and the conviction field $U = \mathsf{T}u$ does not align with $\kappa$: $C$ is central yet value-neutral; $b_2$ is cheap yet strongly held; $b_4$, which feeds $C$, is a conclusion the community wants false. The two fields are $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$, the same operator on two sources, and they are independent vectors.

is the *lock-in parameter*. In the experiments nothing is silenced by hand: the more a channel's reports threaten what the agent values, the less the agent listens. Concretely, each round channel $k$ carries weight $w_k = \exp(-g\,|U \cdot H_k|)$, where $U \cdot H_k$ measures how directly the channel bears on valued commitments and $g$ sets how strongly caring becomes not-listening.

Note what this mechanism *is*: a representation of which conclusions an agent values, expressed as a displacement of where its beliefs come to rest—and, under forgetting, a displacement that saturates (§3.4) while evidence keeps arriving. A bias, not a resistance. The results will make that distinction load-bearing.

### 3.3 Structure learning: reduction and expansion

Structure learning revises $G$ itself—the first requirement of §2, that structure be the *object* of inference, now discharged by two asymmetric operations [10, 26]. The asymmetry is the formal counterpart of normal versus revolutionary science.

*Reduction* is the exact half. Each edge is parameterized by a prior precision $\alpha_e$ ($\alpha_e \to \infty$ pins the edge's parameter to zero; $\alpha_e \to 0$ frees it), the automatic-relevance-determination parameterization of sparse Bayesian learning [20, 31].

Bayesian model reduction [8, 11] scores any pruning in closed form from the already-inverted posterior,

$$\frac{p(o \mid \tilde{M})}{p(o \mid M)} = \mathbb{E}_{p(\theta|o)}\left[\frac{\tilde{p}(\theta)}{p(\theta)}\right], \tag{5}$$

the Savage–Dickey ratio in general form: one inversion scores every candidate reduced structure without re-touching the data.

*Expansion* reaches what reduction cannot: a commitment no one has represented. Following Friston et al. [10], the agent entertains a new node by wiring it tentatively to existing nodes with free priors ($\alpha_e \to 0$) and letting reduction prune the proposal back to the couplings the data support. Expansion is not closed-form—a new node may enlarge the likelihood's support and requires a real inversion—and this cost asymmetry is why a field reorganizes its periphery continually and its centre only rarely. The agent does not plan *when* to expand: candidates arrive at a Poisson rate, and an arrival is kept only if the model log Bayes factor favours the wired-in node (Appendix B). It must be a rate, because one cannot plan toward a dimension one has not yet conceived—Stanford's problem of unconceived alternatives [27], to which we return. The rate itself, however, is not condemned to be a constant: like every precision in active inference it admits treatment as a *belief*—here, the agent's belief that its own model class is incomplete—and Appendix C gives it that treatment, with the constant rate retained as the agnostic baseline every comparison is run against.

### 3.4 Forgetting

The introduction asked how a community loses information it once held; forgetting is the model's answer, and it is load-bearing rather than cosmetic. Read once from fixed precisions, $\kappa = \mathsf{T}\mathbf{1}$ can only grow as learning adds edges: a paradigm with no forgetting is a ratchet, and lock-in under it is automatic rather than earned. We import the standard remedy unmodified: accumulated precision relaxes toward the prior between updates, $\Pi \leftarrow \Pi_0 + \omega(\Pi - \Pi_0)$ with forgetting factor $\omega \in (0, 1]$ [36, 38], so a deposit $k$ steps old carries weight $\omega^k$. The factor is the agent's prior on the world's structural drift—one forgets at the rate one believes the structure is moving [37]. The operator inherits the time index, $\mathsf{T}_t = (I - A_t)^{-1}$, and a once-central node whose couplings go unreinforced relaxes back toward cheapness. With forgetting, the value tilt saturates at $\lambda U/(1 - \omega)$ and competes with the also-saturated per-step evidence, so conviction holds only above a threshold: an entrenched bloc re-tracks a moving truth at low $\lambda$ and stays locked at high—motivated persistence turned from a posited bias into a threshold conviction must clear.

### 3.5 The simulation at a glance

The pieces are now all on the table: the population of §3.1, each agent revising under conviction (§3.2), learning structure (§3.3), and forgetting (§3.4). Concretely, the experiments of §4 instantiate the phlogiston world: a hub-structured paradigm net in which a single core commitment (phlogiston) is load-bearing for combustion, calcination and respiration; at a fixed step $t_{\text{shift}}$ the world's regime flips (the calx is heavier—the gravimetric anomaly), and the truth thereafter contains a dimension (oxygen) that is *not in any agent's network*. One round of the simulation, for every agent:

1. **Observe.** The world emits observations at the belt; channel weights on disconfirming rows are set endogenously by the conviction gate, $w_k = \exp(-g\,|U \cdot H_k|)$ (§3.2), scaled by the agent's gate strength $s$.
2. **Infer.** The agent updates its (value-tilted, Eq. 3) posterior over $x$ by exact linear-Gaussian inference and accumulates precision.
3. **Crisis check (reduction).** Each candidate prune is scored on the move ledger: the closed-form prune evidence $\Delta F$ (Eq. 5) plus $\lambda_i$ times the equally closed-form conviction-value change of the prune, $\Delta U = U^{\top}(\mu_{\text{reduced}} - \mu)$, read from the reduced posterior the Savage–Dickey identity already exhibits. Both protective belt edges scoring $\Delta F + \lambda_i\,\Delta U > 0$ is a *crisis*, and the agent applies the reduction—the hub is released (Appendix B).
4. **Discovery check (expansion).** After reduction, the residual the old paradigm absorbed becomes visible as a coherent pattern in prediction error; when a Poisson proposal arrives (rate $r$), its *direction* is read from that pattern's leading eigenvector, and the wake is accepted iff the model log Bayes factor is positive—the data must hold the new node (§3.3, Appendix B). The rate is constant in the baseline; Appendix C makes it a posterior and a social process, and replaces the one-shot accept with an audited wake-then-prune cycle.
5. **Fuse.** Posteriors and structural edits are pooled across the trust graph, weighted by the intra/inter coupling. Whether agents pool over *all* dimensions or only over those they jointly represent is a protocol choice with consequences (§4.2).
6. **Forget.** Accumulated precision decays by $\omega$ (§3.4).

Crisis, revolution and discovery are thus not stipulated events but threshold crossings of the model's own evidence; Appendix B derives both checks from the single tilted objective and lists explicitly what remains a modelling commitment. Table 1 lists the dials swept; everything else is held fixed, and all code is seeded and ships its own null and positive controls.

One thing deserves emphasis before the results: every ingredient above is its field's default. The belief model is the workhorse of Kalman filtering and structural equations; the value mechanism is the standard formalization of motivated reasoning; the fusion step is classical opinion pooling in information form; expansion and reduction are textbook Bayesian model comparison. Nothing exotic has been smuggled in. The sweep that follows tests what these defaults can do *together*.

| Dial | Symbol | What it controls |
|---|---|---|
| Social coupling | *inter* | inter-community trust weight (0 = isolation) |
| Vanguard fraction | — | fraction of agents in the open (low-$\lambda$) community |
| Conviction | $\lambda_d$ | dogmatic community's value tilt / prune protection |
| Gate scaling | $s$ | strength of the conviction gate on disconfirming channels |
| Proposal rate | $r$ (Poisson) | arrival rate of structural proposals |
| Forgetting | $\omega \in (0, 1]$ | decay of accumulated evidence toward the prior |
| Observation noise | $\sigma_o$ | sensory noise at the belt |
| Fusion protocol | — | pool all dimensions (delta) vs. jointly represented (abstention) (§4.2) |
| Closed-loop mechanisms | $\beta, \tau, \kappa_r, \lambda_c, \epsilon$ | adequacy-driven rate, social excitation, audit credit, value accretion (Appendix |

Table 1: The dials swept in the experiments (2,488 runs); all other quantities are held fixed. The last two rows are protocol and mechanism variations layered on the baseline; each defaults to "off" (constant rate, one-shot accept, delta pooling, static $u$), reproducing the baseline model exactly.

## 4 Results

All results are from the population of §3.1, run by the loop of §3.5 through the full Kuhnian cycle on the phlogiston world—normal science, the gravimetric anomaly, crisis, reduction, discovery of oxygen, new normal, every phase a threshold crossing of the agents' own evidence, none stipulated—swept over the dials of Table 1: 2,488 runs spanning the baseline and every mechanism of Appendix C. One signature organizes everything the sweep found: *conflict resolves into confident compromise.* It is not an accident of the dials. Two classical theorems, one per level of the model, say the sweep could not have come out otherwise.

*Pooling is a contraction.* The fusion step is precision-weighted averaging on a connected trust graph—DeGroot dynamics in information form [60]—and its weights never depend on the *content* of what is pooled. Any such step contracts the spread of beliefs geometrically toward consensus: disagreement can only decay, never regenerate. A persistent school of thought would require an agent to weigh a message by its fit to the agent's own structure, and no quantity in the loop of §3.5 reads that fit; the hazard is recognized within active inference itself [2]. *Gaussian conflict resolves by compromise.* Within an agent the same impossibility holds one level down, and it has been a theorem for fifty years: a posterior built from light-tailed sources responds to conflict between them by confident compromise—it can never reject a conflicting source, however extreme [4, 5]—and its precision *grows* while doing so [6], so contradiction makes a Gaussian agent more certain, never less. There is no slot in $(\Pi, h)$ for being torn: crisis as suspended confidence requires tails the regime does not have.

The two theorems make this community a *null regime*—the simplest faithful model of a believing, communicating community, in which Kuhnian phenomena provably cannot arise. But they are qualitative: what they do not say is where the boundary sits, or how sharply it cuts. That is what the sweep measures, at three levels—within an agent (§4.1), between agents (§4.2), and around the regime (Table 2). The nets are hand-wired to expose the geometry, not fitted to data: nothing here is evidence about chemistry, only about how a structured, bounded, valued mind changes.

### 4.1 Within an agent: conviction tilts where beliefs settle, never when they yield

For a single agent, value buys a biased resting place, never a delayed reckoning [15]: the tilt of Eq. (3) displaces the mean and leaves the update gain untouched (§3.2). The first boundary the sweep locates is what becomes of that result at population scale: the community inherits it exactly, and contact makes it lethal.

*Only total isolation protects a superseded paradigm.* The setup pits a dogmatic community—high conviction, whose agents gate their own disconfirming channels (§3.2)—against an open vanguard already tracking the anomaly, and dials the inter-community trust weight *inter* from 0 (sealed off) to dense contact. Sealed off, the gating protects. But at *inter*$= 0.005$—trust toward outsiders two hundred times weaker than trust within the community—92% convert to oxygen, however strong the gate and however dogmatic the agents, and the transition is a sharp boundary, not a gradual fade (Fig. 3). Contact is corrosive because fusion shares *evidence*, not just conclusions: a small open vanguard's anomalous data flows across the trust channels and sets off each dogmatist's *own* crisis. An agent can silence its own instruments; it cannot silence its neighbours'.

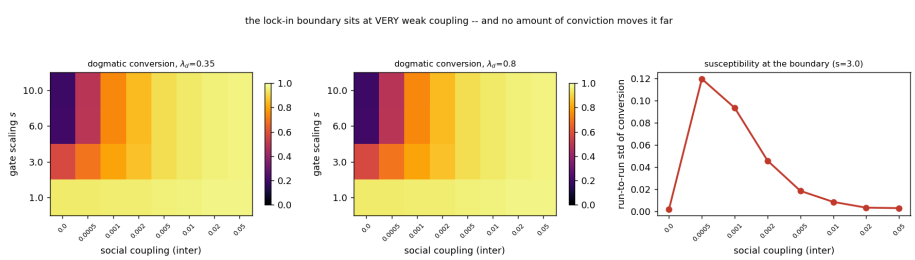



Fig. 3: Only isolation protects a superseded paradigm (phlogiston sweep). *Left, centre:* the fraction of the dogmatic community that converts to oxygen, as a function of contact with the open community (*inter*) and of how strongly conviction gates disconfirming channels ($s$), at two conviction levels ($\lambda_d = 0.35$ and 0.8). Sealed off, gating works: under strong gating ($s = 10$) only 19% ever convert. Past *inter*$\approx 0.005$ nearly everyone converts, and doubling conviction ($\lambda_d : 0.35 \to 0.8$) barely changes the picture. *Right:* the run-to-run spread of outcomes—large exactly where a run can tip either way—spikes in a narrow band around *inter*$\approx 5 \times 10^{-4}$, the signature of a sharp boundary rather than a gradual fade.

*What stalls is the gate—and the gate is the tilt's only route to lock-in.* A control pins down what does the stalling: keep conviction exactly as high but switch the gate off, and the community revolts in full (revolted fraction 1.0, versus 0.0 under gating, $\gamma \to 1$). The evidence was never out of reach; what stalls a community is conviction closing the channels that would deliver it (Eq. 4). So the tilt never stalls anyone, its one causal route to lock-in runs through the gate, and the gate loses to the faintest connection—because a neighbour's evidence arrives weighted by trust, not by the receiver's conviction. Motivated reasoning in this regime is *wishful*, not *dogmatic*: a bias lives in the potential $h$; an entrenchment would have to live in the precision $\Pi$.

### 4.2 Between agents: pooling kills the lone discoverer—by protocol, not by principle

*A discovery made alone does not survive.* Discovery in this model means waking a node—oxygen—that no agent's network yet contains, and since one cannot search for a concept one has not conceived, candidates arrive at random (Poisson rate $r$, §3.3). The discovering itself holds no surprises: the fraction that discovers follows the textbook waiting-time law $1-(1-r)^T$, and the date of the community's *first* discovery is set by the evidence—when the post-reduction residual earns a positive Bayes factor—not by the arrival process. What the rate governs is survival. Under the baseline fusion, a discovery made by one agent at a time is averaged away every time: the moment the discoverer pools beliefs with peers who lack the concept, the shared prior treats the slot the others do not represent as a firmly held "no such thing" and pulls the awakened coupling back to zero. Survival appears only once arrivals cluster the wakings in time—critical mass in *time*, it seems, not in numbers. The next result shows what this law is actually made of.

*Incommensurability is a property of the pooling protocol.* The staggered-discovery death rides on a category error the pooling literature has long flagged [47, 62]: delta pooling reads *has no such variable*—a fact about an agent's model class—as *maximally certain it is zero*, a belief within its support. The corrected protocol is dimension-aware *abstention* pooling: each entry of the precision matrix is pooled only over the neighbours that represent it, while a receiver keeps its own pin—so a concept still cannot arrive through averaging, only through the receiver's own discovery check. The consequence is not a softening but a wholesale relocation of the survival threshold (Fig. 4): under delta pooling staggered survival is zero at every rate; under abstention pooling it tracks the discovery fraction itself. Communities that pool as if absence were denial need their discoverers synchronized; communities whose members abstain on what they cannot represent need only discoverers.

*Every escape goes around the averaging, never through it.* The sweep's remaining mechanisms sharpen the same conclusion (Table 2; full results in Appendix C). None of them lets two connected communities holding the same data *remain* in

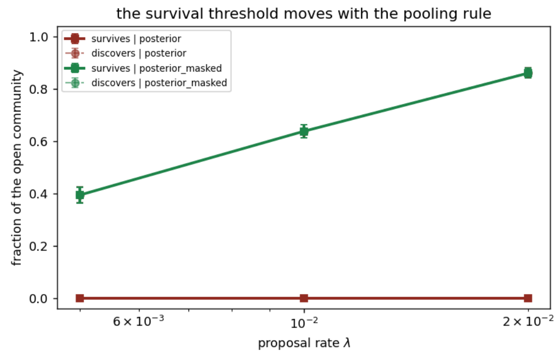



Fig. 4: The survival threshold moves with the pooling protocol (sweep panel F, 32 seeds per cell). Solid: fraction of the open community whose concept survives; dashed: fraction that discovers at all. Under delta pooling (red) a pinned slot is a confident "no such thing" and staggered survival is zero at every rate; under abstention pooling (green)—agents pool each dimension only with neighbours that represent it—survival is 0.39/0.64/0.86 at $r_0 = 0.005/0.01/0.02$, tracking discovery. Incommensurability as a population-level force is a property of the protocol, not of the concepts.

disagreement: each either re-times the passage through the consensus bottleneck or synchronizes the population for it.

| Escape | What it changes | Measured effect | Verdict |
|---|---|---|---|
| Adequacy-inferred rate | *when* the community looks | crisis-to-discovery 55 $\to$ 29 steps; matched by a constant rate of equal mean | re-times the passage |
| Social excitation (Hawkes) | how nearly *together* it looks | waking dispersion 64 $\to$ 2 steps; staggered survival 0 $\to$ 0.88–0.95 | synchrony outruns the averaging |
| Audited wake-then-prune | whether curiosity is *safe* | null world: 100% of wakes re-pinned, exactly | keeps the search honest |
| Value accretion | conviction grown from *inside* | conversion delayed $\sim 2.2\times$, never prevented | buys time, not immunity |

Table 2: Four mechanisms layered on the baseline, none of which opens the consensus engine (Appendix C).

*Robustness.* The arc is robust across forgetting and observation noise ($\omega \in [0.9, 1.0]$, $\sigma_o \in [0.25, 1.0]$): crisis and reduction survive everywhere—including the no-forgetting wall $\omega = 1$, where only discovery thins: a community that never forgets still sees its paradigm die, but loses the capacity to grow the successor. Each experiment ships its own null and positive controls—a world that never shifts produces neither crisis nor discovery; the gated, isolated community stays locked; Poisson discovery follows the waiting-time law—and no acceptance bar is tuned anywhere: the wake's sole accept test is the model log Bayes factor (Appendix B); whatever the derivations there do not fix is named there as a commitment.

## 5 Discussion

The sweep set out to run the Kuhnian cycle and ended by characterizing the regime in which the cycle cannot run; we regard that as the finding. The linear-Gaussian, precision-pooling community is the formalism's *normal-science* limit—smooth, cumulative refinement within a frame, consensus under contact—and nothing in it entrenches, diverges, or breaks. The characterization is of a stylized object—the nets are hand-wired and the dials of Table 1 swept rather than inferred from data—but within it the boundary is measured, not conjectured. What paradigm dynamics require, the measurement locates precisely, and its two nearest requirements are two readings of one word.

*Value as precision.* The conviction field enters the dynamics as a tilt of the potential $h$: it represents *wanting*, and it buys displacement, not defense (§4.1). The dogmatism Kuhn describes—a commitment that resists revision because much is staked on it—is a statement about stiffness, and stiffness lives in $\Pi$: value would have to enter as precision, a convinced agent depositing internal evidence on the commitment it rehearses. The prediction is concrete: lock-in becomes a race between internal rehearsal and pooled external evidence, so the isolation boundary of Fig. 3 should move with rehearsal rate instead of pinning to zero contact. (A subtler channel points the same way: in finite proposal windows, a value-displaced posterior can shape which candidates clear the evidence bar before their window closes—selection on what gets *born*, distinct from resistance on what is held.)

*Trust as inferred precision.* The pooling weights are fixed and content-blind; the conflict-resolution literature's canonical repair is to make each source's precision a latent quantity inferred from its residuals [6], so that a source which keeps disagreeing is discounted rather than averaged in. Within an agent this yields heavy tails, rejection, and falling confidence under contradiction—crisis rendered as a posterior; between agents it is content-gated fusion, under which two communities with enough accumulated credence each discount the other. The prediction is the exact fixed point the contraction theorem forbids: stable schools—connected communities, shared data, persistent disagreement. Both mechanisms stay inside the information-form algebra the operator is built on; both are the immediate future work.

The model's relation to Kuhn can now be stated without allegory. The regime *holds* normal science, with a hard core and protective belt that are derived rather than decreed [66]. It *lacks* the revolutionary remainder: incommensurability within a single precision space [65], crisis as suspended confidence, conversion as a gestalt switch. And it *earns one piece back* in practice rather than concepts: the fusion that crushes a staggered discovery vanishes when agents abstain on dimensions they do not represent (§4.2). The survival result brackets the pooling family at its endpoints—absence read as denial, or full abstention—while real communities presumably sit between; which point they occupy is an empirical question the model converts from a confession into a measurement. Modelling a *proper* Kuhnian revolution means building the exits named above, not reading the cycle into dynamics that provably relax.

Any epistemic community whose aggregation step averages rather than gates inherits the regime's signature, whatever the sophistication of its members. Aggregation by consensus-trained models is precision-weighted pooling at scale: to the extent that machine-mediated science inserts such aggregators into its loop, it will refine frames and never grow one. The model makes this testable; it does not yet establish it. The same statement is a foundation for multi-agent active inference at large: a population of structure learners is only as *open-ended* as its pooling permits, because mean-field sharing forecloses exactly the dimensions that expansion exists to add. Open-endedness is a property of the protocol, to be designed in—not an emergent grace of scale.

The deepest exit lies beyond the present model class. Everything revolutionary above passes through model expansion, and every candidate edits the prior over a fixed likelihood—while a genuinely new *instrument* enlarges the likelihood's own support. Structure learning over latent structure is closed-loop; structure learning over the sensorium is not, and we know of no agnostic stand-in for it as clean as the Poisson rate was for unconceived latents. This is the formal face of the limit Stanford named: one cannot place a posterior over the unconceived [27], only over whether one's model has missed something—enough to set how hard an agent looks, never where (Appendix C). What a community controls is how hard it looks—and, these results say above all, what its pooling does to the member who finds it first.

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## A The operator, the tilt, and the Schur residue

*Linear-Gaussian construction.* The couplings form a structural equation model $x = Ax + \zeta$, $\zeta \sim \mathcal{N}(0, D^{-1})$ with $A$ the weighted DAG adjacency [18]; the choice is for tractability, and a nonlinear coupling would replace $\mathsf{T}$ by a local linearization. The propagation operator is the resolvent

$$\mathsf{T} = (I - A)^{-1} = \sum_{k \geq 0} A^k, \qquad \mathsf{T}_{iw} = \sum_{\text{paths } i \rightsquigarrow w} \prod \pi \;=\; \pi_{i \rightsquigarrow w}, \tag{6}$$

the Neumann series finite because a DAG makes $A$ nilpotent. We realise $\mathsf{T}$ on the coupling magnitudes $|A|$ so downstream mass does not cancel when a child opposes a parent; the signed $A$ builds the precision matrix. We index $\mathsf{T}$ so that $\kappa_i$ sums row $i$; under the SEM convention the released code stores the transpose and computes the same fields as column-sums of $\mathsf{T}^\top$, so a reader reproducing $\mathsf{T}\mathbf{1}$ literally against the code's array must transpose first.

*The value tilt.* With honest posterior $p(x \mid o, G) = \mathcal{N}(\mu, \Sigma)$, the tilted posterior of Eq. (3) is a pure mean shift, $q_\lambda(x) = \mathcal{N}(\mu + \lambda \Sigma U, \Sigma)$, with cumulant-generating function $Z(\lambda) = \lambda U^\top \mu + \frac{1}{2}\lambda^2 U^\top \Sigma U$. Note that gating a channel ($\rho_k \to 0$) *raises* the posterior variance and hence $Z''$; lock-in lives in the vanishing mean-update gain $\rho_k H_k \Sigma U$, not in the curvature.

*The Schur residue.* Marginalizing a hub condenses its influence into new edges among its neighbours: for a common-cause structure over $(x_1, x_2, b)$,

$$\Pi = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{marginalize } b} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \tfrac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1\ 1) = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{pmatrix}, \tag{7}$$

a direct coupling induced where the joint had a zero. Conditioning on a shared *effect* is the dual operation: two commitments that jointly feed one observation are a priori independent, and clamping that observation anti-couples them—explaining away, run on the precision matrix. Both operations fill in the off-diagonal coupling the carry-over cost is built from, and Fig. 5 draws both on the smallest nets that exhibit them.

## B The crisis and discovery checks, derived from the ledger

The per-round loop of §3.5 contains two threshold events: the *crisis check* (when does an agent prune the protective edges and release the hub?) and the *discovery check* (when does it wake the unconceived node?). This appendix states exactly what each check computes, derives both from the single objective the paper already carries, and lists the modelling commitments that remain after the derivation. The aim is that nothing in the two checks should read as a stipulated event: every acceptance is the model's own evidence weighed on the model's own ledger, and everything that is *not* derived is named as a commitment.

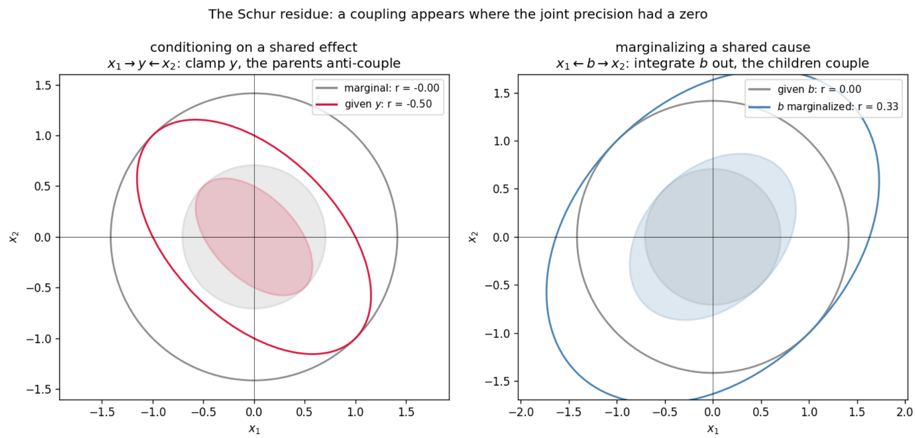


Fig. 5: The Schur residue, computed on the smallest nets that show it. *Left:* conditioning on a shared effect ($x_1 \to y \leftarrow x_2$): the parents are a priori independent (grey, $r = 0$); clamping their common descendant $y$ manufactures an anti-correlation (crimson, $r = -0.50$)—explaining away. *Right:* marginalizing a shared cause (the matrix of Eq. 7): the children are independent given $b$ (grey); integrating $b$ out leaves them coupled (blue, $r = +0.33$). In both cases a coupling appears where the joint precision had a zero—the off-diagonal mass the carry-over cost $\kappa = \mathsf{T1}$ is built from.

### B.1 One ledger for every structural move

The bounded agent of §3.2 revises its beliefs by minimizing $D_{\mathrm{KL}}[q \,\|\, p(x \mid o, G)] - \lambda\, \mathbb{E}_q\big[U^\top x\big]$ (Eq. 3): fidelity to the honest posterior, traded against the value of where the commitments land, at temperature $\lambda$. A structural move—pruning an edge, waking a node—changes both terms, so the same objective scores the move:

$$\mathrm{score}(M \to \tilde{M}) \;=\; \underbrace{\Delta F}_{\text{log Bayes factor}} \;+\; \underbrace{\Delta G}_{\text{epistemic gain}} \;+\lambda\; \underbrace{\Delta U}_{\text{conviction value}}, \qquad \text{accept} \iff \mathrm{score} > 0, \tag{8}$$

with $\Delta F$ the evidence for the edited structure, $\Delta G$ the expected information gain about any newly introduced latent (zero for a prune), and $\Delta U = \mathbb{E}_{q_{\tilde{M}}}\big[U^\top x\big] - \mathbb{E}_{q_M}\big[U^\top x\big]$ the change in the conviction-field value of the commitments. Equation (8) is not a new mechanism: it is Eq. (3) evaluated at two structures and differenced. Both threshold events of the simulation are instances of it.

### B.2 The crisis check is the ledger on a prune

For a prune, every term of Eq. (8) is closed-form in the linear-Gaussian setting. The evidence half is Bayesian model reduction (Eq. 5): writing the agent's accumulated likelihood deposit as $J = \Pi - \Pi_0$, $j = h - h_0$, the Savage–Dickey identity scores the swap of the prior $(\Pi_0, h_0)$ for the edge-$e$-reduced prior $(\Pi_0^{(e)}, h_0^{(e)})$ under the same likelihood, and—this is the part the main text does not spell out—it *already exhibits the reduced posterior*,

$$\Pi^{(e)} = \Pi_0^{(e)} + J, \qquad h^{(e)} = h_0^{(e)} + j, \tag{9}$$

so the value half costs one linear solve per candidate edge:

$$\Delta U_e \;=\; U^\top\big(\mu^{(e)} - \mu\big), \qquad \mu^{(e)} = \big(\Pi^{(e)}\big)^{-1} h^{(e)}, \quad \mu = \Pi^{-1} h, \tag{10}$$

with $U = \mathsf{T}u$ the agent's *current* conviction field, read along its present couplings. The crisis check is then exactly the ledger: edge $e$ is flagged iff $\Delta F_e + \lambda\,\Delta U_e > 0$, and *crisis* is the round at which both protective belt edges are flagged, upon which the agent adopts the reduced prior composed over every edge its own ledger flags (the hub released). Pruning a conviction-protective edge moves the posterior mean away from the cherished commitments, so $\Delta U_e < 0$ and the rule reproduces the intuitive form "the evidence must overcome $\lambda$ times the value the edge protects."

A natural surrogate suggests itself here: the static threshold $\Delta F_e > \lambda\, v_e$ with $v_e = |U_{\text{parent}}| + |U_{\text{child}}|$, the summed conviction magnitude at the edge's endpoints—available without a solve, and with the right *form* (it is Eq. (8) with $\Delta U_e \equiv -v_e$). But it has the wrong *content*: $v_e$ is a static stand-in for a quantity the model can compute exactly, so we compute it. The difference is not cosmetic, and its direction is itself informative: because the reduced and full posteriors share the likelihood deposit, the mean displacement of a prune—and with it $\lambda\,\Delta U_e$—*shrinks as evidence accumulates.* A data-rich paradigm is barely protected at the threshold; almost all of its persistence must come from upstream, from the conviction gate that decides which evidence is allowed to accumulate at all (Eq. 4). The static surrogate would quietly overstate threshold protection; the exact ledger locates lock-in where the paper's results put it: gating plus isolation, not a high bar at the moment of decision.

### B.3 The discovery check is the ledger on an expansion

Expansion cannot be closed-form (§3.3): a new node needs a real inversion, and—Stanford's problem of unconceived alternatives—the agent cannot plan toward a dimension it has not conceived. The discovery check therefore has exactly two parts, and only two:

1. **Arrival.** Candidate proposals arrive at a Poisson rate $r$. This is the one place a stipulated process enters, and it is a deliberate modelling commitment: a rate is the maximally agnostic model of conceiving what one has not conceived. The proposal's *direction* is read from the model's own prediction errors: a hidden hub $b$ wired to the visible nodes with couplings $c$ leaves, on marginalization, the rank-one footprint $-cc^\top/\Pi_{bb}$, so the leading eigenvector of the windowed error second moment is the best single-hub explanation of the residual. The window is an estimator horizon, not a gate.

2. **Acceptance.** The arrived proposal is accepted iff the *model log Bayes factor* of the bordered (hub-wired) model against the agent's current one is positive—that is, iff the residual genuinely loads on the proposed hub, equivalently iff the immediate Savage–Dickey prune of the new couplings would be rejected. The epistemic gain $\Delta G$ sits on the ledger too, but on the other side of the move: being positive for *any* new latent, it is what pays for attempting the (in general expensive) inversion, and it cannot keep a hub the data do not hold—an accept test that included it would wake on pure noise. Entertaining is paid by curiosity; keeping is paid by evidence. One consequence is owned rather than hidden: repeated looks give the Bayes factor a small false-discovery rate (rare noise crossings of $\Delta F > 0$ in the null world). We regard this as a feature of the honest model—communities do occasionally adopt spurious structure—and its principled remedy is not a reinstated threshold but the wake-then-prune cycle named in the limitations, in which a spuriously woken node is pruned back by the same reduction that scored it.

A tempting third part is deliberately absent: a *trigger*, requiring the residual floor (the eigenvalue above) to clear some height, perhaps for a sustained number of steps, before the Bayes factor is consulted. Such a gate would duplicate the accept test's job with a free constant—the Bayes factor already rejects proposals the residual does not support—and its only principled reading, a budget on when to pay for the expensive inversion, is a statement about real scientific communities rather than about this simulation, where the inversion is a small log-determinant. The discovery check therefore has no tuned threshold: the Poisson rate carries Stanford's commitment, the eigenvector carries the direction, and the ledger decides. (The floor is still recorded as telemetry, because it is the visible signature of the residual the old paradigm had been absorbing.)

### B.4 What remains a modelling commitment

The derivation above leaves a short, explicit list of choices that are *not* forced by the formalism. We state them because the paper's claim that "crisis, revolution and discovery are threshold crossings of the model's own evidence" is honest only if its boundary is drawn:

- **The belt conjunction.** "Crisis" is operationalized as *both* named mass-law edges flagged in the same round. The per-edge accept is derived; the conjunction over those two edges is our operational definition of having eliminated the phlogiston mass-law reading.
- **The ordering — a result, not a commitment.** The simulation attempts expansion only after an agent's own crisis, but the ordering is enforced by the evidence rather than by fiat: running the identical proposal + ledger pipeline for every pre-crisis agent at every step (without applying it), the ledger rejects 13,180 of 13,185 pre-crisis agent-steps; the five crossings are marginal ($\Delta F < 0.005$, two orders below the post-crisis acceptance evidence)

and are exactly the repeated-look false-positive rate owned above, whose remedy is the wake-then-prune cycle (§B.5). The intact paradigm absorbs the residual; there is nothing for the check to find until reduction exposes it.
- **The Poisson rate** $r$ (swept), the prediction-error *window* (an estimator horizon), a short *warm-up* before reduction is permitted (an initialization guard), and the released hub's self-precision are quantitative choices of the same standing as the other dials of Table 1.
- **The closed-loop mechanisms' own commitments.** The mechanisms of Appendix C replace the constant rate, the one-shot accept and the static $u$ with inferred quantities, but import quantitative choices of their own: the Hawkes kernel's parametric form ($\beta$, $\tau$, exponential memory)—the *reading* of the kernel as filtered social evidence is motivated below, its shape is chosen; the expiring credit's geometric decay $\lambda_c$, a fixed-form surrogate for recomputing $\Delta G$ on the current posterior; and the audit's grace and patience horizons. All are Table 1-class dials, swept or defaulted, never tuned against an outcome.

### B.5 The rate, made principled (and the audit that pays its debts)

This subsection records the inferential readings that license the closed-loop mechanisms whose quantitative results Appendix C reports: the adequacy-driven rate, the social excitation, the audit that the full ledger's acceptance rule makes necessary, and the accretion dynamics behind the Lakatos result—together with what each leaves stipulated.

*The adequacy belief, filtered.* One explores at the rate one believes one's model is incomplete, and that belief has a licensed likelihood: the residual floor the discovery check already records—the prediction error that persists after every within-model update, whose persistence is therefore evidence about the model *class* rather than its parameters. The agent tracks it with a fixed-gain filter, $s \leftarrow (1-\rho)\, s + \rho \cdot \text{floor}$, the same exponential-forgetting form as $\omega$ (and the fixed-gain limit of conjugate Gamma–Poisson updating under a drift prior on the latent rate), and sets $r(t) = r_0 + \kappa_r \max(s - s_{\text{ref}}, 0)$, capped at $r_{\max}$. A persistent residual raises the agent's own arrival rate; a quiet model lets it decay to the agnostic base. The clean waiting-time law of §4.2 does, as anticipated, dissolve into a feedback system; the experiments therefore report realized (effective) rates, and the matched-mean control of Appendix C is what separates "rate when needed" from "more rate."

*The social stream, and why its filter is a Hawkes kernel.* Because the agents share one hidden world, model inadequacy has a common cause across the community, and a trusted neighbour's discovery is evidence about one's *own* adequacy variable—the situation in which federated inference prescribes pooling [47]. Filtering the neighbours' discovery events (a point process, observed through the trust weights $T_{ij}$ the population already fuses over) under an exponentially-correlated prior on the latent inadequacy yields a posterior intensity that is a shot-noise process: a baseline plus a decaying bump per observed event. That is exactly the linear self-exciting form of Hawkes [12],

$$r_i(t) \;=\; r_0 \;+\; \sum_{j\in\partial i} T_{ij} \sum_{t_j<t} \beta\, e^{-(t-t_j)/\tau}, \tag{11}$$

with $t_j$ the neighbours' discovery times: $\beta$ the trust-scaled evidential weight of one observed discovery, $\tau$ the assumed correlation time of inadequacy. The structural-proposal channel the model already has is thereby promoted from carrying *content* (which node to wake) to also carrying *momentum* (how eagerly to look)—a formal home for the observation that revolutions arrive in waves—and the branching ratio becomes the model's measurable answer to how much social excitement a revolution needs (Appendix C, Fig. 9). What the reading does not discharge: the kernel's parametric form is chosen, and the excitation deposited by a wake that is later pruned back (below) is not retracted—the community's excitement is an observation of the wake, not of the verdict.

*The wake-then-prune audit.* Accepting wakes on the full ledger $\Delta F+\Delta G > 0$ lets curiosity adopt structure the data do not (yet) hold, so the acceptance rule itself creates the obligation to audit. The audit re-asks the wake's own question every round: the bordered-model log Bayes factor of the agent's *current* (conditioned) beliefs against its *pre-wake* anchor at the stored coupling —not a Savage–Dickey prune of the wired prior, which would compare two priors and find the hub favoured by construction, since the slot has no observation row and is held only by the data's mean displacement along its pattern. The $k$-th audit's verdict is

$$\text{verdict}_k \;=\; \Delta F_{\text{now}} \;+\; \Delta G\, \lambda_c^{\,k}, \tag{12}$$

the wake's ledger with *expiring* credit: epistemic gain pays for entertaining, but it is a diminishing resource—information gain about a latent shrinks as the posterior over it sharpens—so the geometric $\lambda_c$ is a fixed-form surrogate whose principled successor is recomputing $\Delta G$ on the current hub posterior. A patience-length run of non-positive verdicts (after a grace period covering post-wake consolidation) re-pins the slot to its exact dormant form, and the agent may wake it again later: entertaining is paid by expiring curiosity, keeping by evidence. Figure 6 shows the cycle doing the false-discovery work the commitments list above assigns it: in a null world that never shifts, the credit wakes hubs in every agent and the audit re-pins every one of them; in the flip world the genuine hub is woken everywhere and never pruned.

*Value accretion (the Lakatos closure).* The conviction source $u$, exogenous and static in the baseline, gains slow dynamics: per node, $u_{\text{eff}} \;=\; u\,(1+g)$ with $g \leftarrow \text{clip}(g+\epsilon\, e-\delta\, g,\, 0,\, g_{\max})$ and $e$ the agent's own entrenchment (the diagonal of its accumulated evidence deposit $\Pi-\Pi_0$, max-normalized)— self-evidencing made slow [13]: value flows toward the commitments the programme's own data have entrenched, and through $U \;=\; \mathsf{T}u$ and the gate of Eq. (4) a maturing programme increasingly silences its own disconfirming channels. The clip bounds

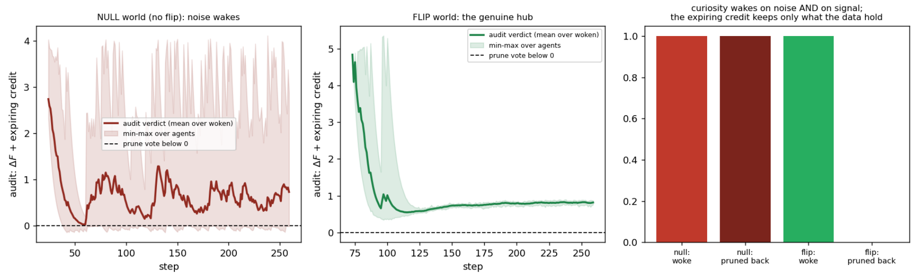


Fig. 6: The wake-then-prune cycle controls the repeated-look false-discovery rate with no threshold. *Left:* null world (no regime shift)—the audit verdict (evidence plus expiring credit) of every noise-woken hub decays below zero and the slot is re-pinned exactly. *Centre:* flip world—the genuine hub's verdict stays positive; the data take over before the credit runs out. *Right:* curiosity wakes structure in both worlds; the audit keeps only what the data hold.

the accretion–gate feedback loop. Figure 7 shows the resulting phase picture: gain accreting over the run, the realized gate climbing with $\epsilon$, and conversion at the lock-in boundary falling—progressive and degenerating problemshifts as dynamics (Appendix C).

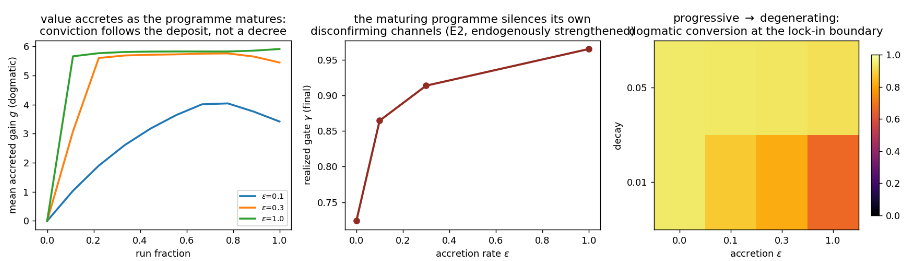


Fig. 7: Dynamic conviction (Lakatos, run forward). *Left:* the accreted gain $g$ grows as the programme matures—conviction follows the agent's own evidence deposit, not a decree. *Centre:* the realized self-silencing gate $\gamma$ strengthens with the accretion rate $\epsilon$. *Right:* dogmatic-community conversion at the lock-in boundary over the ($\epsilon$, decay) plane: a programme that matures into conservatism delays its supersession—and still loses to connection.

## C Closed-loop mechanisms: quantitative results

The baseline model holds four quantities fixed that admit treatment as beliefs or protocols: the proposal rate, the acceptance of a wake, the conviction source $u$, and the pooling rule. The inferential readings that license each mechanism are derived in Appendix B.5; this appendix reports what each does to the population results. Every mechanism defaults to "off"—constant rate, one-shot accept, static $u$, delta pooling—reproducing the baseline exactly, and each comparison below is run against that baseline with its own control.

*The rate as a posterior over model adequacy.* The agent tracks its residual floor—the leading eigenvalue of the windowed prediction-error second moment, the error no within-model update can absorb—with a fixed-gain filter $s \leftarrow (1-\rho)\, s + \rho \cdot \text{floor}$ and reads its proposal rate off the posterior, $r(t) = r_0 + \kappa_r \max(s - s_{\text{ref}}, 0)$. The result (Fig. 8) has the signature a posterior should have and a dial cannot: flat at $r_0$ through normal science, surging after the agent's own crisis exposes the residual (to an effective 0.34 from $r_0 = 0.01$), and decaying once the woken node explains it. The crisis-to-discovery delay falls from 55 to 29 steps—but the control that matters is the fixed rate it *matches*: a constant-rate agent given the adaptive run's realized mean rate achieves the same delay (27 steps). The gain is not more exploration; it is exploration when the agent's own evidence calls for it. Stanford's constraint survives: only the *propensity to look* is inferred, never a direction.

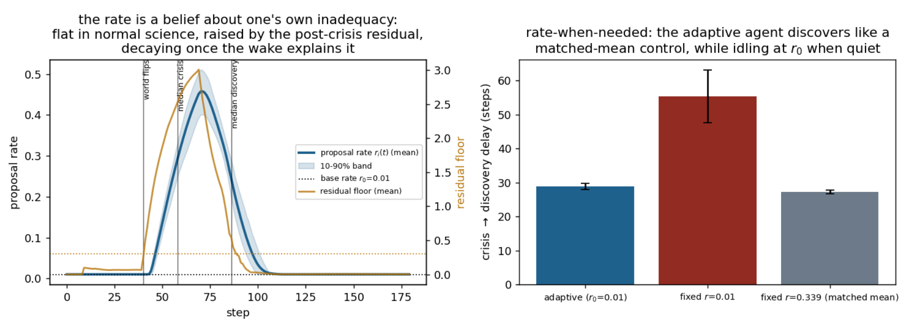


Fig. 8: The proposal rate as a posterior over model adequacy. *Left:* the per-agent rate (blue, mean and 10–90% band) is flat at the base rate $r_0$ through normal science, rises with the post-crisis residual floor (amber, right axis)—the evidence of model-class inadequacy the within-model updates cannot absorb—and decays once the woken node explains it. *Right:* crisis-to-discovery delay. The adaptive agent (left bar) matches a fixed-rate control given its realized mean rate (right bar) while idling at $r_0$ when the model is adequate: the gain is rate-when-needed, not more rate.

*Social excitation rescues the staggered discovery.* Filtering neighbours' discovery events through the trust weights, under an exponentially-correlated prior on the latent inadequacy, yields a posterior intensity of exactly the linear self-exciting (Hawkes) form [12],

$$r_i(t) \;=\; r_0 \;+\; \sum_{j\in\partial i} T_{ij} \sum_{t_j<t} \beta\, e^{-(t-t_j)/\tau}, \tag{13}$$

with $t_j$ the neighbours' discovery times, $\beta$ the trust-scaled evidential weight of one observed discovery, and $\tau$ the assumed correlation time of inadequacy (Appendix B.5). Under Eq. (13) the first discovery raises the whole neighbourhood's propensity to look; the cascade compresses the community's wakings from a dispersion of 64 steps to 2, and—because survival is *completeness*-driven (a single still-pinned peer holds the fused slot precision enormous for every neighbour)—the completed cascade lets the anchored couplings consolidate. Survival rises monotonically in $\beta$ from exactly zero at $\beta = 0$ to 0.88–0.95 at $\beta = 1.5$ across base rates $r_0 \in \{0.005, 0.01, 0.02\}$ at which an unexcited community essentially never completes (Fig. 9; 32 seeds per cell). The branching ratio of Eq. (13) is the model's quantitative answer to how much social excitement a revolution needs. One qualifier: survival is completeness-driven and therefore horizon-bound—a constant-rate community given a high enough rate and a long enough horizon also completes its cascade; what excitation adds is the mechanism by which a community reaches near-simultaneity at rates serendipity alone essentially never does.

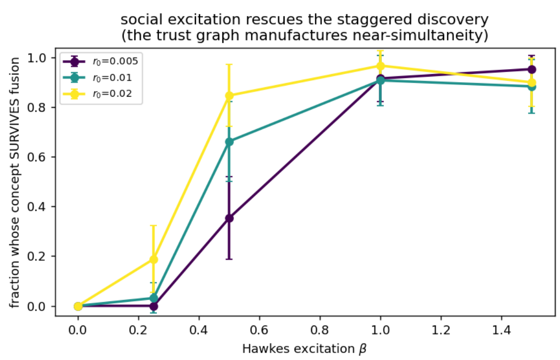


Fig. 9: Social excitation rescues the staggered discovery (sweep panel E, 32 seeds per cell). Fraction of the open community whose oxygen concept survives fusion, versus the excitation $\beta$ of Eq. (13), at three base rates. At $\beta = 0$ (the constant rate) survival is exactly zero: staggered wakings are crushed one at a time. Excitation compresses the community's wakings in time, the cascade completes, no pinned peer remains in the fusion average, and survival rises to 0.88–0.95—the near-simultaneity that survival requires, manufactured by the community itself.

*The audited wake-then-prune cycle.* Accepting wakes on the full ledger $\Delta F + \Delta G > 0$ lets curiosity adopt structure the data do not yet hold, and the audit of Appendix B.5 pays that debt: in a null world that never shifts, curiosity wakes structure in 100% of agents and the audit re-pins 100% of those wakes, exactly; in the flip world the genuine hub is woken everywhere and never pruned (Fig. 6). Two companion measurements close two more commitments. Widening the search—three eigen-hub candidates, each with its own dormant slot, plus the three largest residual couplings, 1,140 candidates priced on the same ledger—does not dilute discovery: the true cause wins 53 of 60 acceptances (median cosine 0.999 against the rank-one baseline). And running the identical discovery check *before* each agent's crisis, at every step, the ledger rejects 13,180 of 13,185 pre-crisis agent-steps: the crisis-before-expansion ordering is enforced by the evidence, not by fiat—the intact paradigm absorbs the residual, so there is nothing for the check to find until reduction exposes it.

*Value accretion (Lakatos, run forward).* Letting $u$ accrete onto the commitments the agent's own evidence deposit has entrenched (Appendix B.5) closes the loop through the gate of Eq. (4): a maturing programme increasingly silences its own disconfirming channels ($\gamma$ climbs from 0.72 to 0.97 as $\epsilon$ rises), conversion at the lock-in boundary falls from 0.94 to 0.67, and at very weak coupling the dogmatic community's conversion is *delayed* roughly 2.2-fold rather than prevented (Fig. 7). Conviction that grows from the inside still loses to connection: it buys the old paradigm time, not immunity—a fair gloss of a degenerating problemshift.

*The mechanisms compose.* With the adequacy-posterior rate, social excitation, the expiring-credit audit, abstention pooling and value accretion all enabled in one population, the full arc still runs—crisis at $t \approx 67$, discovery at $t \approx 69$, the rate surging $0.02 \to 0.77$ and decaying back to base, zero false prune-backs, concept survival complete, and the never-shifting null world silent in every channel. No mechanism breaks another, because none adds a new objective: each routes a quantity the ledger already computes—residual, epistemic gain, evidence, deposit—back into a parameter that had been a constant.